

THESIS PRELIMINARY OUTLINE

Goal document

(max. 2 pages)

Title: Developing a Cost-Effective Digital Twin Framework for Energy-Efficient Operation in Existing Buildings

VINAY DILIP JAYASELAN | vinay.dilip_j.6534@student.lu.se | JAN 2026 – JUNE 2026

Aim: To develop and evaluate a cost-effective digital twin framework for existing buildings by integrating low-cost sensors and data-inference techniques, and to assess its ability to support accurate energy monitoring and energy-efficient operational decision-making at a zone level.

Introduction

Background and Motivation

Digital twins offer strong potential for improving building performance, but most existing buildings lack the extensive, high-quality sensor infrastructure required for traditional implementations. Retrofitting older buildings is often too costly, limiting access to this technology. Low-cost sensors provide an affordable alternative, but their data is often noisy, incomplete, or unreliable. This creates a need for digital twin approaches that can operate under real-world constraints.

This thesis explores a solution by combining low-cost sensing with data-inference methods to create a functional, affordable digital twin for a single room or zone. The goal is to demonstrate that useful energy insights can be generated even with imperfect data, offering a practical pathway for improving energy efficiency in older buildings.

Why This Topic Was Chosen

Most existing buildings cannot support traditional digital twins due to limited sensors and high upgrade costs. Low-cost sensors and inference methods now offer a realistic way to bring digital twin capabilities to older buildings. This study aims to develop an affordable, practical approach that can improve energy efficiency where it is needed most.

Hypothesis and Objectives

These hypotheses evaluate whether a low-cost digital twin can be both technically accurate and practically useful for energy efficiency.

Hypotheses

- **H1: Accuracy**

The digital twin developed using low-cost sensors and inference methods will estimate HVAC, lighting, and plug-load energy use with an acceptable level of accuracy compared to reference or benchmark data.

- **H2: Data Improvement**

Data-inference techniques (such as occupancy estimation, plug-load prediction, and missing-value reconstruction) will significantly improve the completeness and reliability of the low-cost sensor data.

- **H3: Practical Use**

The digital twin will be able to identify energy-wasting patterns and provide actionable insights that support energy-efficient operation in existing buildings.

The overall goal is to create and test a low-cost digital twin prototype for older buildings. This involves combining affordable sensors with smart data methods and evaluating if the system is both accurate and useful.

Objectives

- **O1: Data and Sensing**

To identify and select a set of low-cost sensors capable of capturing key parameters related to HVAC, lighting, and plug-load energy use in existing buildings.

- **O2: Data Inference and Processing**

To develop data-inference methods (e.g., occupancy estimation, plug-load prediction, missing-value reconstruction) that compensate for sparse or noisy sensor data.

- **O3: Digital Twin Development**

To design and construct a zone-level digital twin prototype that integrates low-cost sensing and data-inference techniques into a functional modeling framework.

- **O4: Accuracy Evaluation**

To evaluate the accuracy of the digital twin in estimating HVAC, lighting, and plug-load energy use compared to reference or benchmark data.

- **O5: Energy-Efficiency Insights**

To assess the extent to which the digital twin can provide actionable insights that support energy-efficient operation in existing buildings.

Research Methods and Approach

This section explains the step-by-step plan for this project.

THESIS PRELIMINARY OUTLINE

Goal document

(max. 2 pages)

1. Research Design

A technical study involving framework design, prototype development, and performance evaluation.

2. Data Collection

Collect key parameters such as temperature, humidity, CO₂, light levels, plug-load electricity use, and basic weather data using low-cost sensors or synthetic datasets when needed.

3. Data Processing and Inference

Clean sensor data, correct errors, and apply inference methods (e.g., occupancy estimation, plug-load prediction, missing-value reconstruction) to enhance data quality.

4. Digital Twin Development

- Build a simplified room-level energy model using EnergyPlus or a Python-based model.
- Integrate processed sensor data with the model to form the digital twin prototype.

5. Testing and Evaluation

- **Accuracy:** Compare estimated energy use with benchmark data using metrics such as MAE & RMSE.
- **Usefulness:** Assess the system's ability to detect inefficiencies (e.g., unnecessary plug loads, heating/cooling anomalies).

6. Scalability Discussion

Discuss how the prototype could be extended to multi-zone or whole-building applications and identify limitations of low-cost sensing.

7. Reporting

Document the framework, methods, results, and conclusions in the thesis.

Scope and Limitations

Scope

- The study develops a digital twin for a single room or thermal zone, rather than an entire building.
- It uses low-cost sensors to measure key parameters such as temperature, light levels, and plug-load electricity use.
- The digital twin includes a simplified energy model, data-processing methods, and inference techniques to handle missing or noisy data.
- The system is evaluated based on its accuracy in estimating energy use and its ability to provide useful energy-efficiency insights.
- When real building data is unavailable, synthetic or simulated data will be used to test the framework.

Limitations

- The study is limited to one zone, so interactions between multiple rooms or building-wide dynamics are not considered.
- Low-cost sensors may introduce noise, drift, or measurement errors, which can affect the accuracy of the digital twin.
- The simulation models rely on simplified assumptions about occupancy and equipment behavior, which may not fully reflect real-world conditions.
- The digital twin is not deployed in a live building; instead, it is tested using offline datasets, meaning real-world energy savings are not measured.

The study does not include economic analysis or business considerations; the focus is strictly on technical development and evaluation.

Expected Results

Results: The study is expected to show that a functional digital twin can be built using low-cost sensors and inference methods. The prototype should provide reasonably accurate estimates of HVAC, lighting, and plug-load energy use and demonstrate that inference techniques significantly improve data quality. The digital twin is also expected to identify operational inefficiencies and offer useful energy-saving insights, showing potential for practical application in existing buildings.

Contributions

- A practical framework for developing low-cost digital twins in existing buildings.
- Demonstration of how inference methods can compensate for sparse or noisy sensor data.
- A scalable approach that can begin at room level and expand to whole buildings.

Support for making digital twin technology more accessible and affordable.

Required Resources and Previous Knowledge

Low-cost sensors, a basic computer, Python, and a simple building-simulation tool. Foundational knowledge in building physics, data processing, programming, and energy-efficiency concepts is required

Role of the Different Parties

THESIS PRELIMINARY OUTLINE

Goal document

(max. 2 pages)

Examiner

- Checks academic quality, scope, and required knowledge.

Supervisor

- Guides the research, methods, and structure.

Co-supervisor

- Provides technical or domain-specific support (if assigned).

Student

- Plans and performs the work, builds the prototype, runs tests, and writes the thesis.

Date & Signatures:

Examiner: **Victor Fransson**

Main Supervisor: **Pieter De Wilde**

Second Supervisor: **Ilia Iarkov**

Student: **Vinay Dilip Jayaselan**